

BIOMECHANICS (EPHE 341)

Projectile motion

McGinnis Chpt 2 pg 62–73

Projectile Motion

A **projectile** is any object, not in contact with the ground, that is allowed to move under the influence of gravity and no other external force.

- the acceleration of the object is constant (uniform acceleration)
- The human body can be considered a projectile whenever it is airborne (running, jumping, falling, etc.)

Because everything on earth has the same acceleration due to gravity we can use equations to predict where an object that is in flight will be if we know certain variables

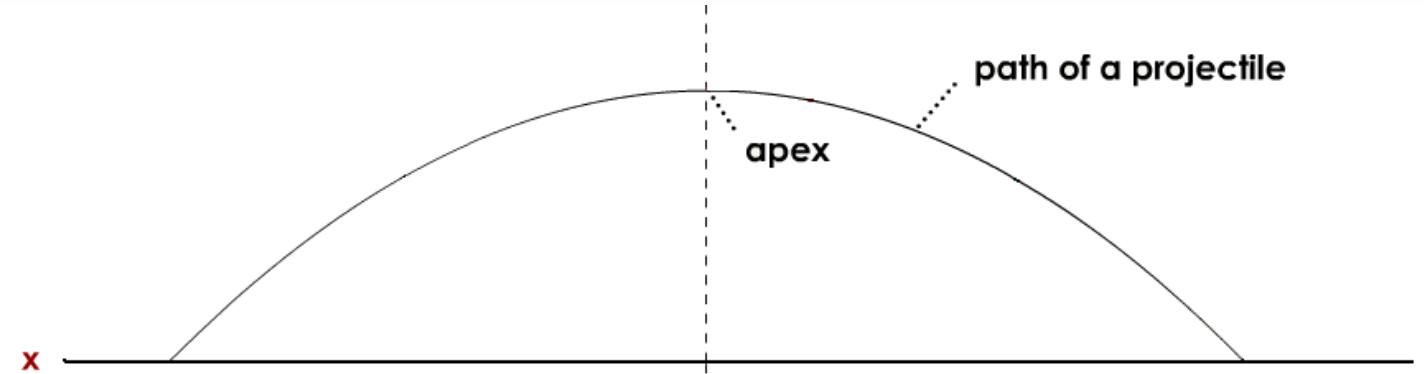


Projectile Motion



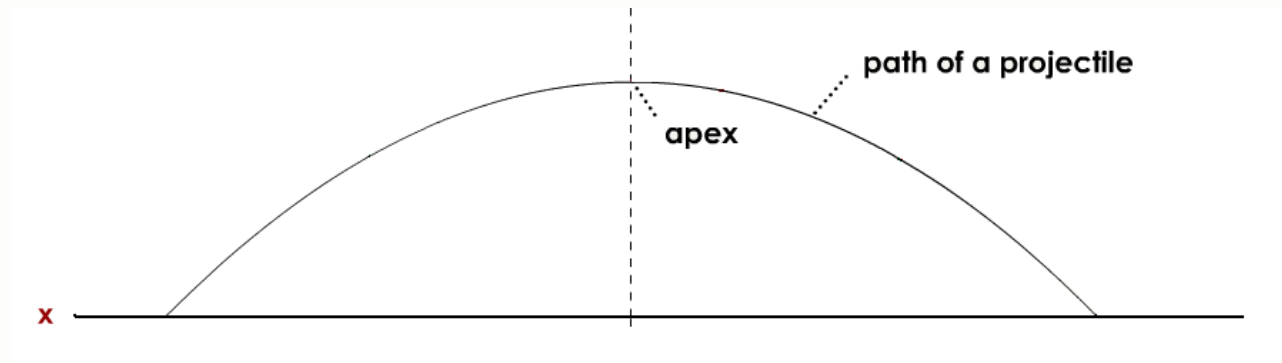
Projectile Motion

- A parabola is a **symmetrical** curve.
- That is, it has a peak (also called **apex**) in the middle and the curve is the same on both sides of the peak

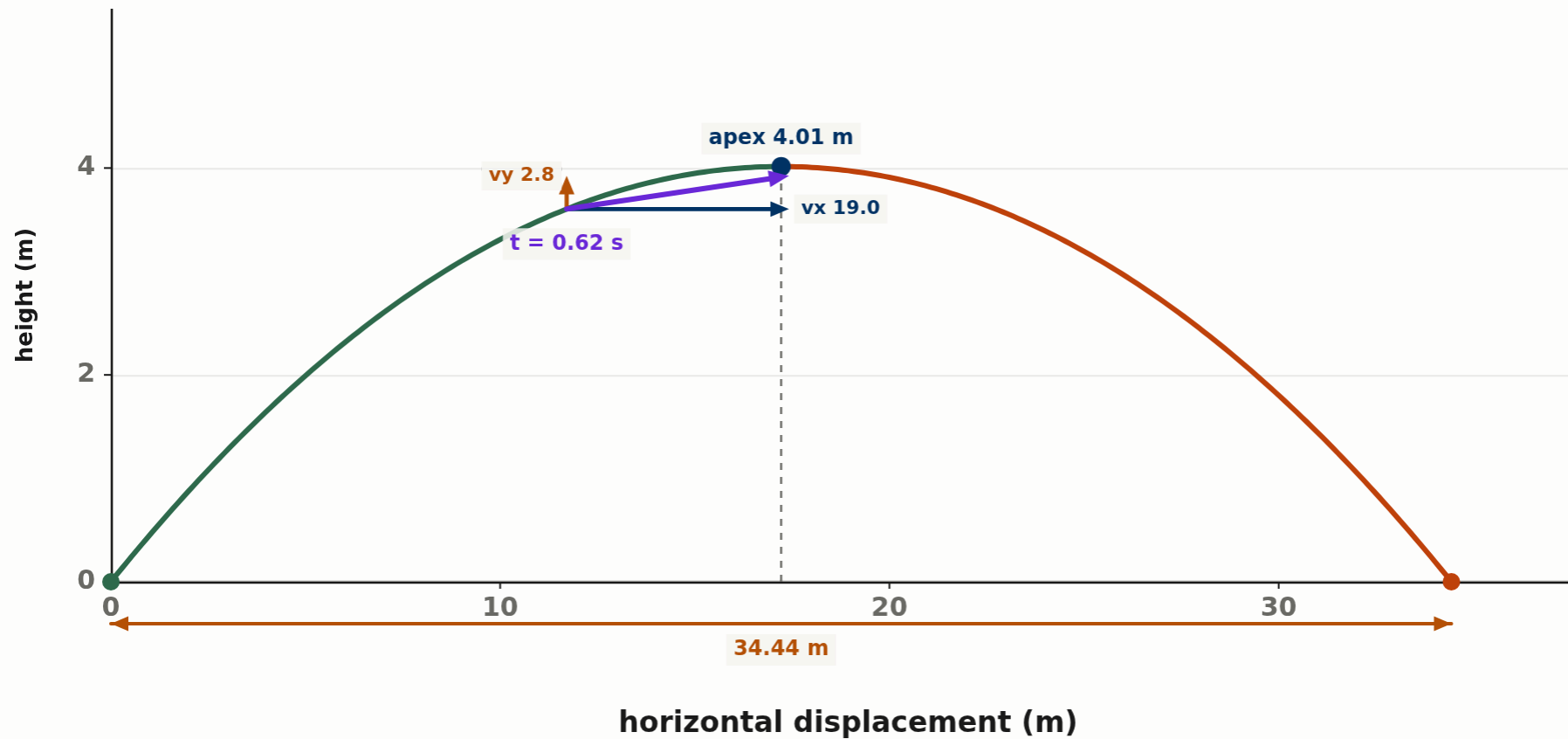


Why a parabola?

- Why do objects on earth follow a parabolic path when thrown?
- We need to think about force vectors and derive equations of motion
- We know that when we throw an object it will follow a path like the one below. Assume that the blue vector is the initial velocity.
- When the object is released or thrown it has only one force of gravity acting on it.



The two components, separately



$v_x = 19.03$ m/s (constant) · $v_y = 2.84$ m/s · flight 1.81 s · apex 4.01 m · range 34.44 m

The horizontal arrow never changes length — there is no horizontal force once the object leaves the ground. The vertical arrow shrinks, reverses at the apex and grows again. Adding them at every instant is what bends the path into a parabola. With the launch and the landing level, the apex sits at exactly half the flight time.

soccer ball: 21 m/s at 25°

baseball: 33.3 m/s at 37°

a jump: 6 m/s at 45°

straight up: 29.4 m/s at 89°

- Consider the x (horizontal) and y (vertical) components **separately**.
- Suppose a ball was thrown straight up and the vertical displacement was determined at different times.
- **How would we find out the initial velocity if we knew that it took 6 seconds for the object to leave the ground and then land?**

Some important questions

- if an object is thrown vertically upward?
- What would be the equation to describe it's acceleration?
- If you have the equation for acceleration how would you determine it's equation for velocity and displacement?
(Hint: use calculus)

Same chain as the calculus lecture: differentiate to go down, integrate to come back up.

Deriving the vertical equations

- Due to gravity the vertical acceleration of a projectile is -9.81 m/s^2
- Therefore we can make an equation for acceleration for the vertical motion of any object in free flight

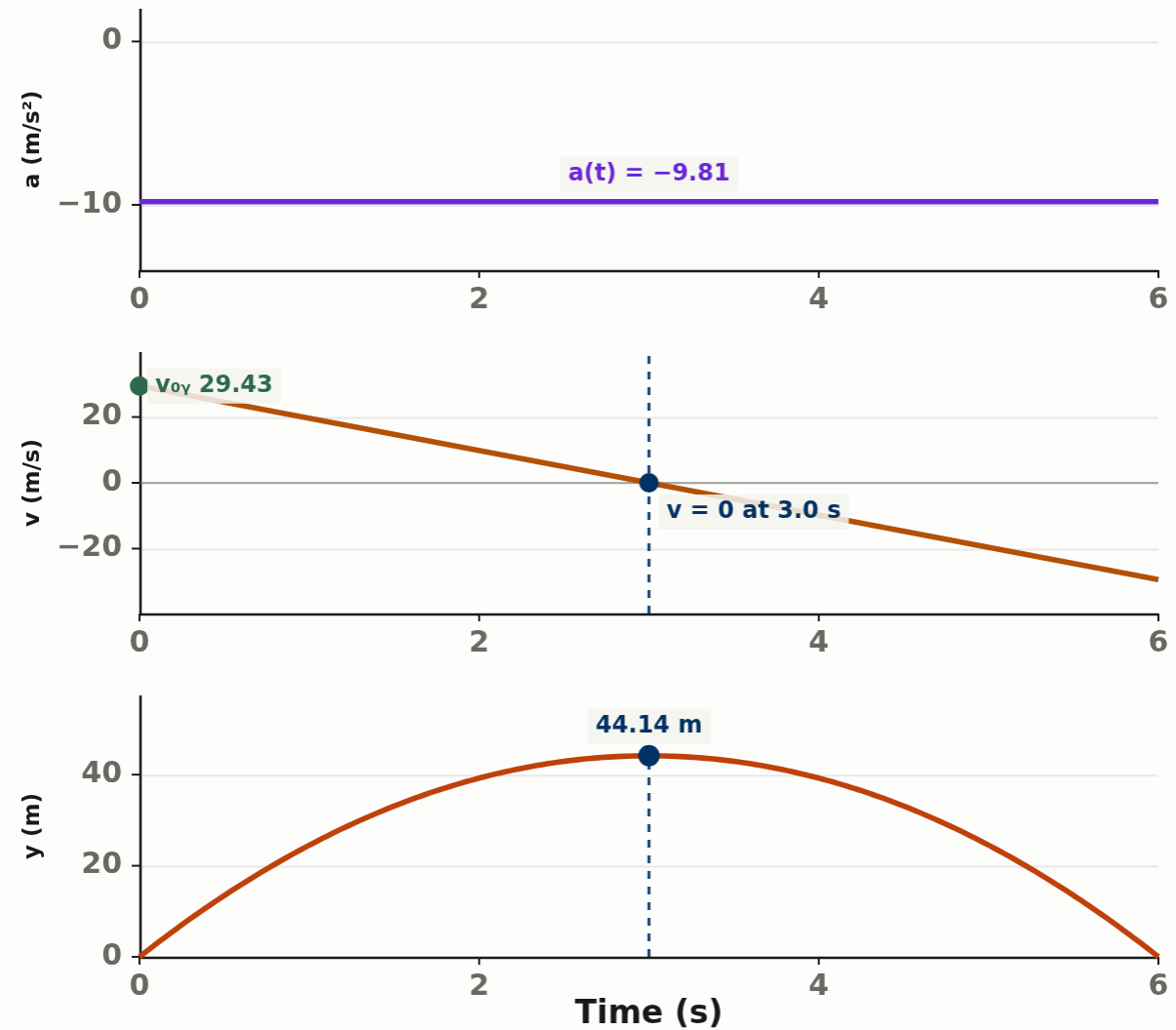
$$a_y(t) = -9.81$$

- to find the equation for velocity we need to take the **integral**. When we do this we get the equation

$$v_y(t) = -9.81t + v_{oy}$$

- to find the equation for displacement we find the integral again

$$d_y(t) = (-9.81/2)t^2 + v_{oy}t + d_{oy}$$



flight time 6.0 s

We add unknowns to represent the initial values

Back to the question

- For an object thrown vertically how would we find out the initial velocity if we knew that it took 6 seconds for the object to leave the ground and then land?
- We derived the equations of motion, lets look at them to see which can help us solve this problem
- **Which equation(s) would you use to solve this problem? What information do you need to know?**

Which equation?

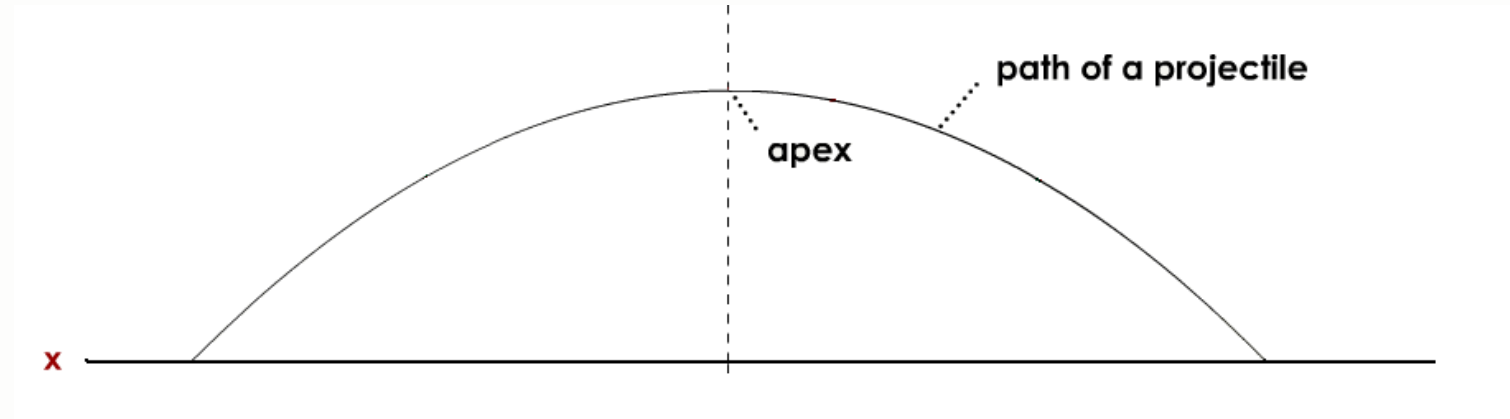
- Since we are looking for initial velocity the equation below would be the simplest one to solve

$$v_y(t) = -9.81t + v_{oy}$$

- But we only know t and so we need to find out what $v_y(t)$ is.
- Maybe some information about projectiles may help us solve the problem

Important features of projectiles

- At the **apex the vertical velocity is zero**
- Also the apex is the highest point achieved
- If a projectile takes off and lands at the same height then the apex occurs at $\frac{1}{2}$ **the total flight time**
- The vertical component of velocity at the beginning of the flight is equal but opposite to the vertical component at the end of the flight



Solving for the initial velocity

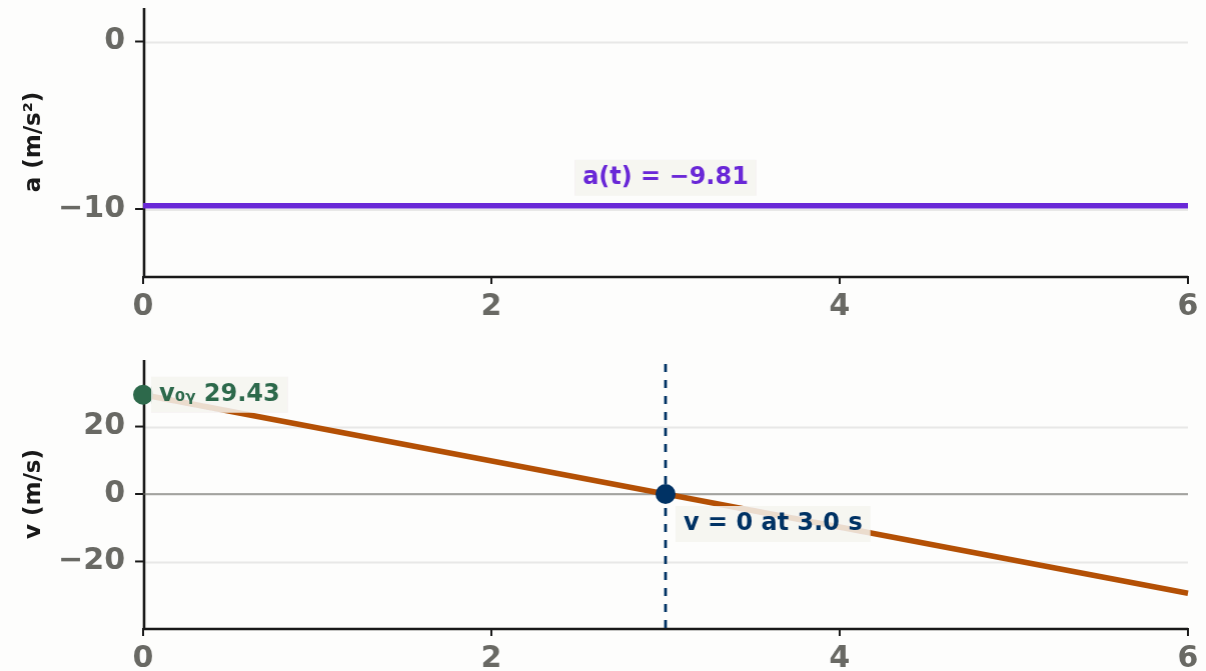
- So now we can use that information to solve for the initial velocity... how?
- if we solve for the following equation at the apex then we know $t = \frac{1}{2}(6 \text{ s}) = 3 \text{ s}$ and $v_y(t) = 0 \text{ m/s}$

$$v_y(t) = -9.81t + v_{oy}$$

$$0 = -9.81(3) + v_{oy}$$

$$v_{oy} = \mathbf{29.43 \text{ m/s}}$$

That leaves us with one unknown... the initial velocity



flight time 6.0 s

And the maximum height

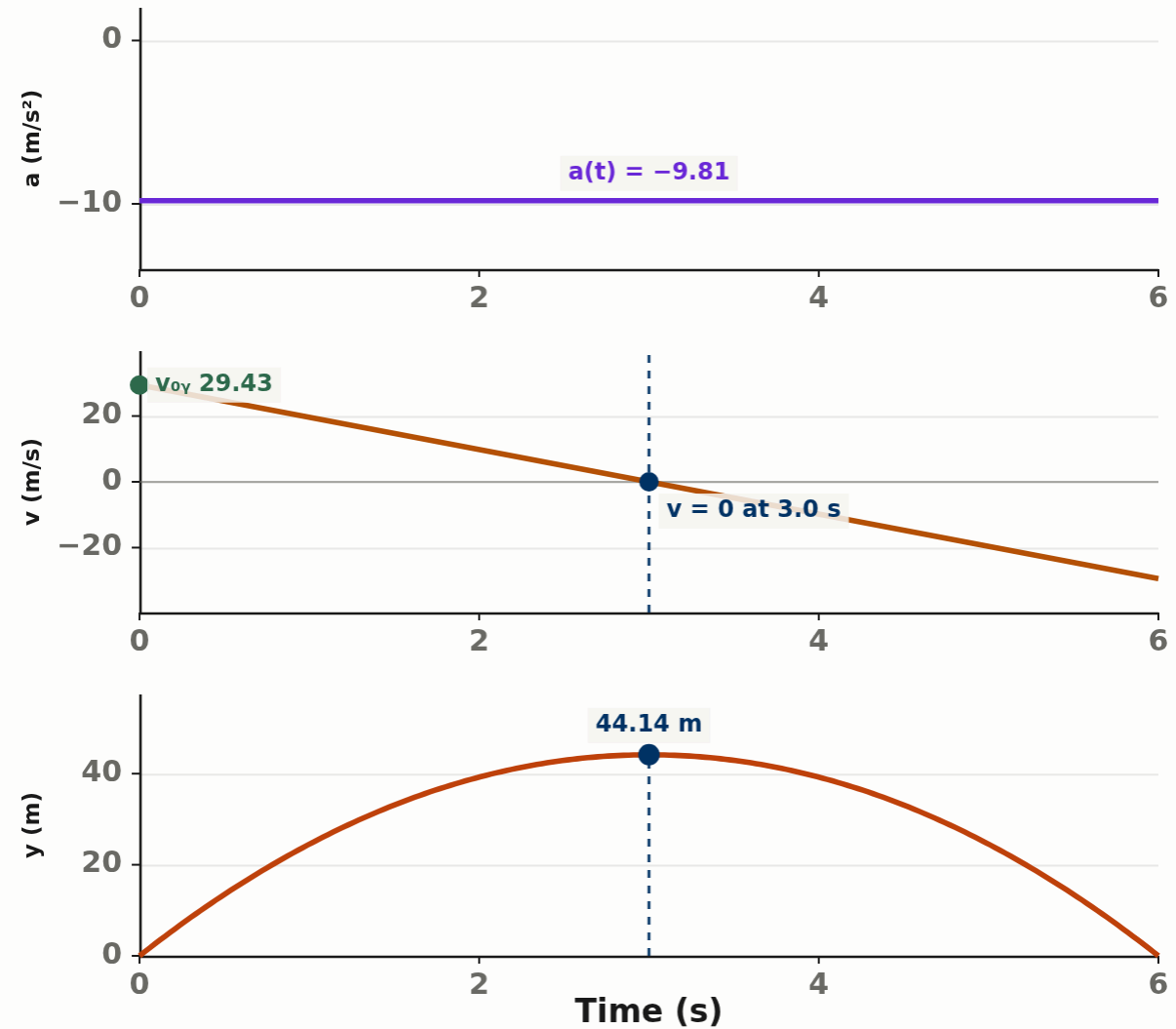
- Now we can find some other values as well, such as what is the maximum height achieved
- To find the highest vertical displacement we need to solve for the $d_y(t)$ at the apex
- We know t at the apex is $\frac{1}{2}(6) = 3$ s and if we assume $d_{oy} = 0$

$$d_y(t) = (-9.81/2)t^2 + v_{oy}t + d_{oy}$$

$$d_y(3) = (-9.81/2)(3)^2 + 29.43(3) + 0$$

$$d_y(3) = -44.145 + 88.29$$

$$d_y(3) = \mathbf{44.145\text{ m}}$$



flight time 6.0 s

- Now that we understand the vertical acceleration, velocity and displacement of a projectile lets talk about the **horizontal** kinematics
- When an object is not in contact with the ground it's horizontal acceleration is **zero**
- Remember that the acceleration due to gravity only acts vertically downward

$$a_x(t) = 0$$

Deriving the horizontal equations

- Then we can determine the equations for horizontal velocity and displacement

$$a_x(t) = 0$$

to find the equation for velocity we need to take the integral. When we do this we get the equation

$$v_x(t) = v_{ox}$$

to find the equation for displacement we find the integral

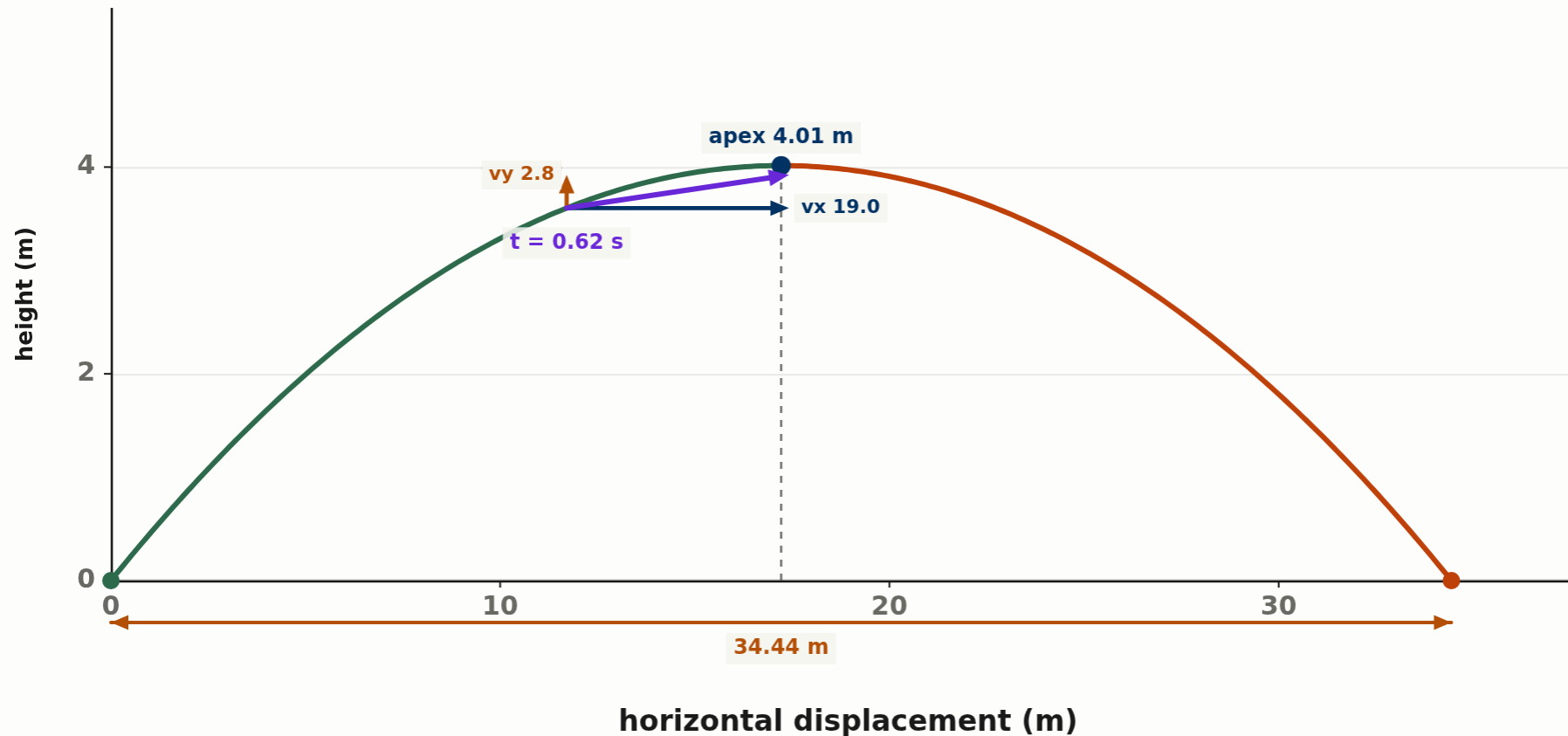
$$d_x(t) = v_{ox}t + d_{ox}$$

Why are horizontal projectile kinematics important?

- The horizontal velocity stays constant throughout the flight. This results in a horizontal displacement defined by **the time that it takes the object to rise and fall vertically**
- It is both the vertical and horizontal kinematics of a projectile that defines the parabolic flight path.

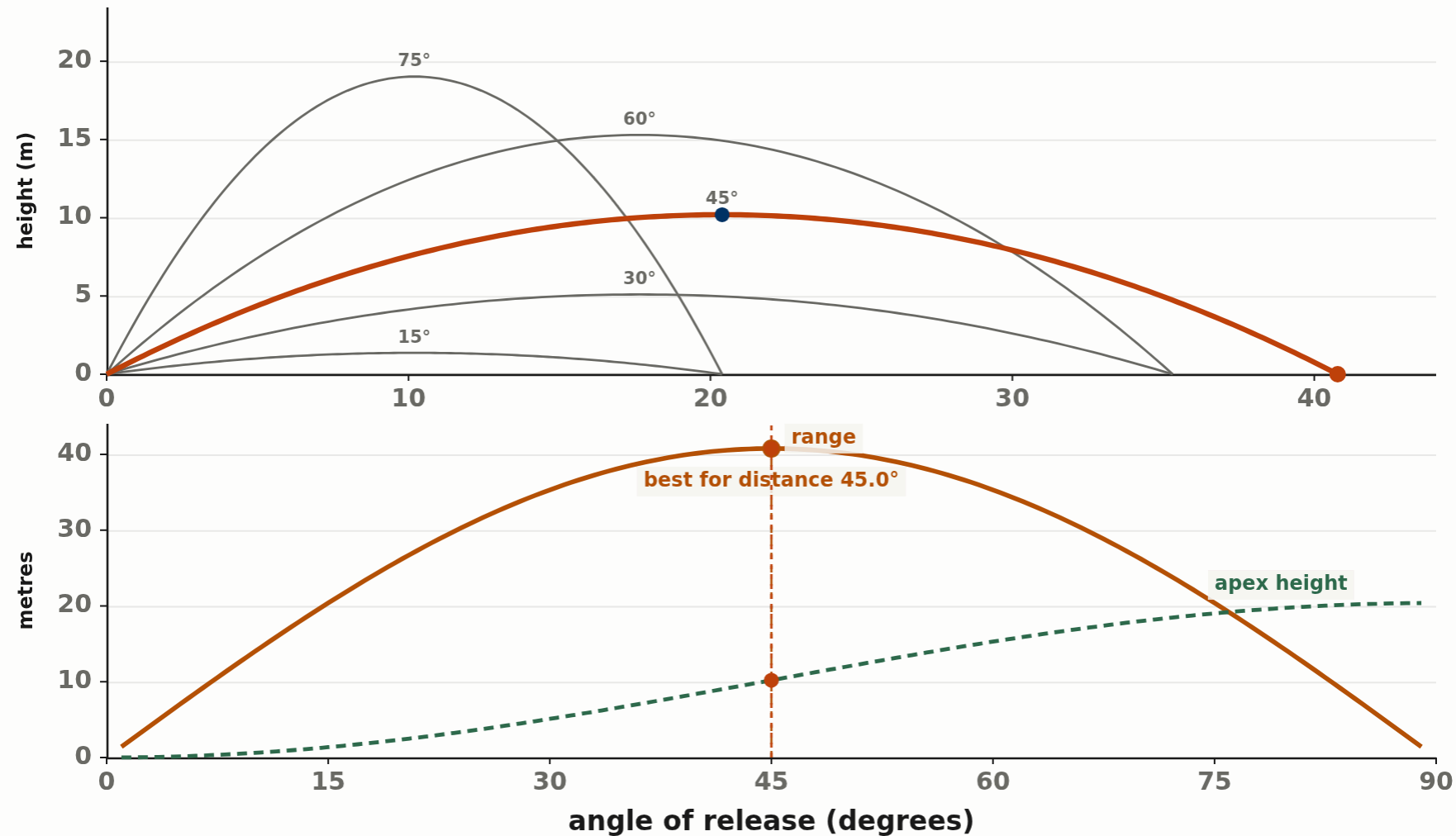
Adding the components back together

If we consider the horizontal and vertical velocity vector components together they demonstrate why the object has a parabolic path. If we add the components we can see that the resultant velocity vector is in the direction of the parabolic flight (as we would expect)



The angle of release

45° for distance, 90° for height



at **45°**: range **40.77 m**, apex **10.19 m** · best angle for distance **45.0°**

From the ground the range curve peaks at **45°**, and it is symmetric about it — 30° and 60° carry exactly the same distance. Height keeps climbing all the way to 90°, because every degree you add goes into the vertical component.

Two more equations

- I always prefer to use the derived equations of motion but here are two more equations that you can use to solve projectile motion problems
- **But... where did they come from?**

$$d = \frac{1}{2}(v_0 + v_f)t$$

$$v_f^2 = v_0^2 + 2ad$$

Where $d = \frac{1}{2}(\mathbf{v}_o + \mathbf{v}_f)t$ comes from

- Let's see if we can find out where this equation came from. It is similar to the equation for displacement

$$d_y(t) = \frac{1}{2} at^2 + v_{oy}t + d_{oy}$$

If we assume that $d_{oy} = 0$

$$d_y(t) = \frac{1}{2} at(t) + v_{oy}t$$

$a = v/t$, or $v = at$, or $v_f - v_o = at$

$$d_y(t) = \frac{1}{2} (v_f - v_o)t + v_{oy}t$$

$$2d_y(t) - 2v_{oy}t = (v_f - v_o)t$$

$$2d_y(t) = v_ft - v_ot + 2v_{oy}t$$

$$d_y(t) = \frac{1}{2} (\mathbf{v}_{oy} + \mathbf{v}_{fy})t$$

Where $v_f^2 = v_o^2 + 2ad$ comes from

- Let's see if we can find out where this equation came from. Let's start with 2 equations that we already know

$$v_y(t) = at + v_{oy}$$

$$d_y(t) = \frac{1}{2} at^2 + v_{oy}t + d_{oy}$$

And solve for t in the first one

$$v_{fy} = at + v_{oy}$$

$$t = (v_{fy} - v_{oy})/a$$

Then sub in the equation for t into the second equation

Where $v_f^2 = v_o^2 + 2ad$ comes from

$$d_y(t) = \frac{1}{2} at^2 + v_{oy}t + d_{oy} \quad \text{and} \quad t = (v_{fy} - v_{oy})/a$$

$$d_{yf} = \frac{1}{2} a[(v_{fy} - v_{oy})/a]^2 + v_{oy}[(v_{fy} - v_{oy})/a] + d_{oy}$$

$$d_{yf} = \frac{1}{2} (v_{fy}^2 - 2v_{fy}v_{oy} + v_{oy}^2)/a + (v_{oy}v_{fy} - v_{oy}^2)/a + d_{oy}$$

$$2d_{yf} - 2(v_{oy}v_{fy} - v_{oy}^2)/a - 2d_{oy} = (v_{fy}^2 - 2v_{fy}v_{oy} + v_{oy}^2)/a$$

$$2d_{yf} - 2d_{oy} = (v_{fy}^2 - 2v_{fy}v_{oy} + v_{oy}^2 + 2v_{oy}v_{fy} - 2v_{oy}^2)/a$$

$$2ad_{yf} - 2ad_{oy} = v_{fy}^2 - v_{oy}^2$$

Therefore to use this equation we need to assume that d_{oy} is zero

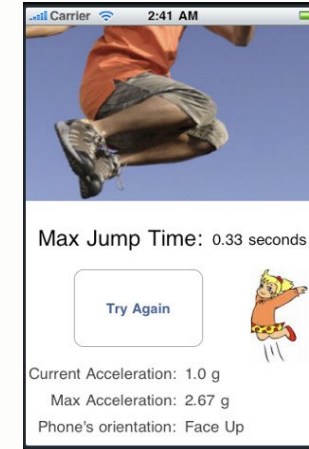
Measuring flight time

Jump mat

- Measures time in the air and outputs jump height
- So then what calculations does the jump mat use to determine jump height?
- What could be some errors involved in the measurements?

Force plate — measures time in the air and calculates initial velocity

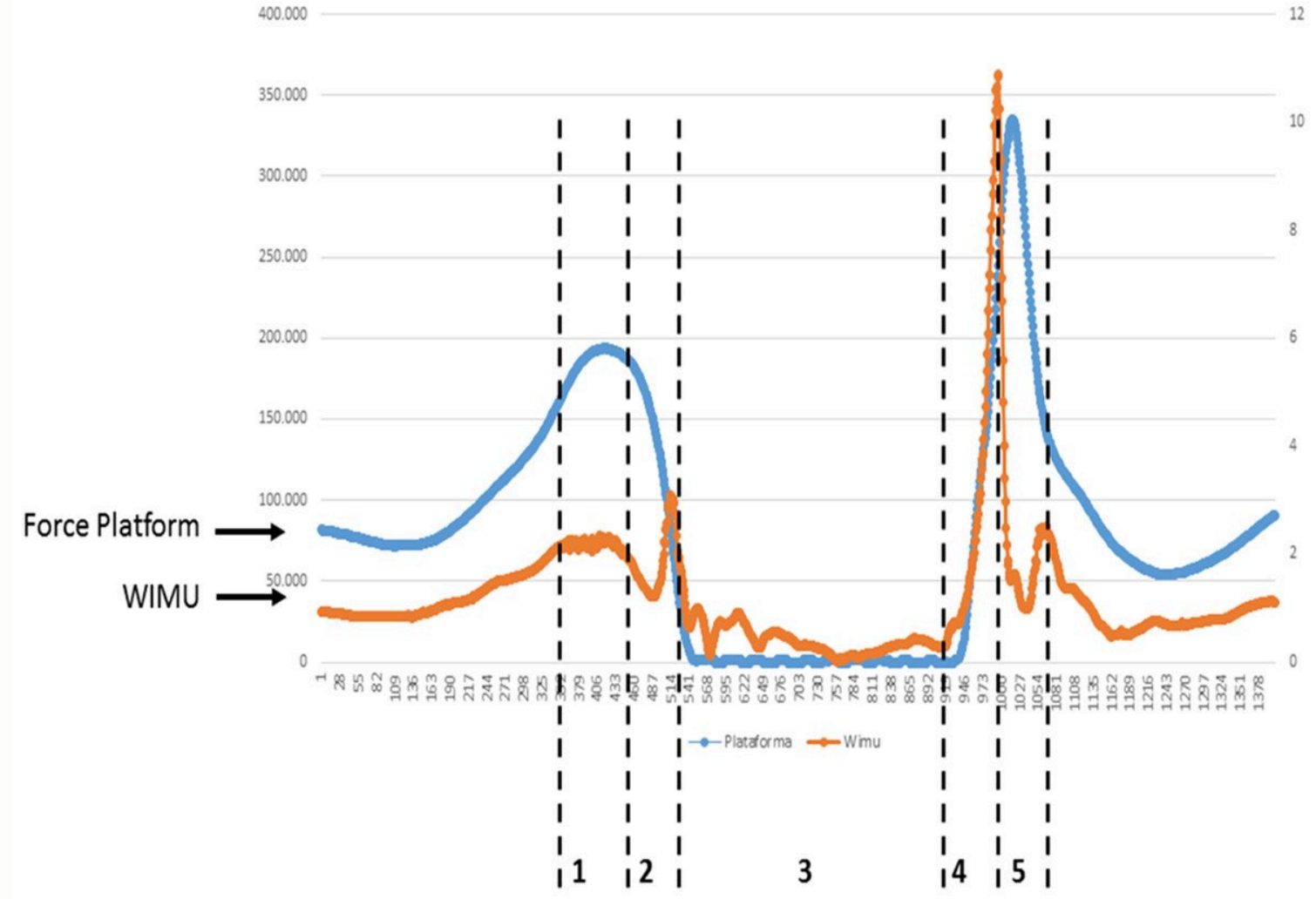
iPod — jump time app; measures the period of time when the accelerometer doesn't measure any force



Reading the flight off the signal

iPod acceleration data, from “accelerometer data logger”. The flat section is the flight: **0.6 s** with no force on the accelerometer at all.

Force platform and wearable inertial unit recorded together.



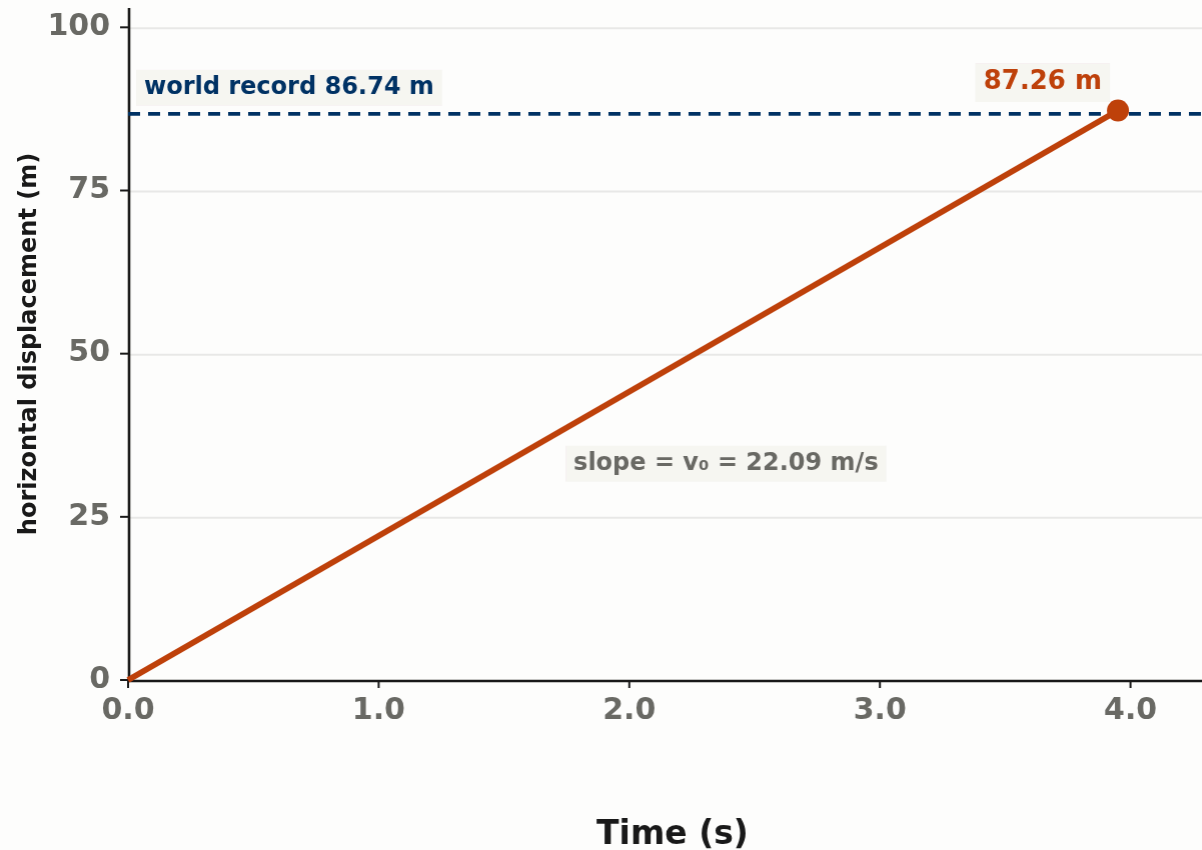
Strategies to Solving Projectile Problems

1. Determine specifically what is asked for and when necessary draw a diagram
2. List all known variables — include those given in the question and those that you are able to figure out without calculation
3. Derive the equations of motion and choose the equation that allows you to use your variables to solve for the answer to the question (this may take more than one step as you may have to solve for other unknowns first)
4. Plug in your variables and solve for the answer
5. Do a reality check (ie: check to see if your values seem reasonable)

Example Problem #1



Example Problem #1 — the figure



Olympic hammer, 22.09 m/s for 3.95 s

1 · what we are given

2 · the horizontal displacement equation

3 · against the record

world record 86.74 m

Yes, he breaks the world record by **0.52 m** with a throw of 87.26 m

$d = v_0 t = \mathbf{87.26\ m}$ · record 86.74 m · **breaks it by 0.52 m**

Nothing vertical enters this at all. Horizontal acceleration is zero, so horizontal displacement is just velocity multiplied by the time the hammer happened to spend in the air.

1 · what we are given

2 · the horizontal displacement equation

3 · against the record

Example Problem #2

A high-jumper is attempting to break the women's world high jump record of 2.09 m (6'10"). If she has a vertical component of initial velocity of 6.16 m/s, does she break the world record (or at least achieve a height that would allow her to clear the bar)?



Solution

- In this question, we are given: V_0 (vertical) = 6.16 m/s
- Since we are dealing with a vertical velocity, we know that acceleration is equal to gravity: $a = -9.81 \text{ m/s}^2$
- We are only considering the upward path of the high jumper, therefore, the final vertical velocity (at the apex of the jump) will be zero: V_f (vertical) = 0 m/s

$$v_y(t) = -9.81t + v_{oy}$$

$$0 = -9.81t + 6.16 \rightarrow t = 0.629 \text{ s}$$

$$d_y(t) = -(\frac{1}{2})9.81t^2 + v_{oy}t + d_{oy}$$

$$d_y(0.629) = -4.905(0.629)^2 + 6.16(0.629)$$



Solution (alternative way)

- Alternatively you could use another equation; for example we could use:

$$v_f^2 = v_i^2 + 2ad$$

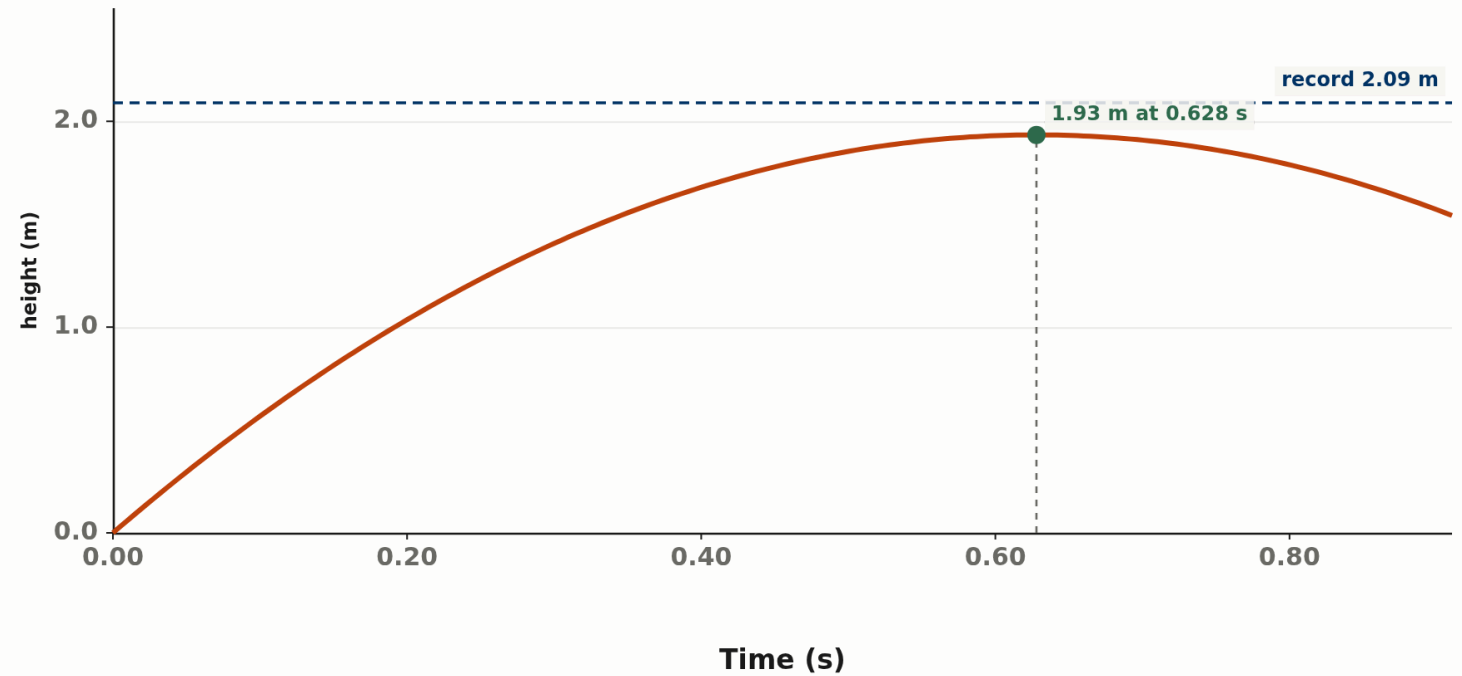
$$(0 \text{ m/s})^2 = (6.16 \text{ m/s})^2 + (2)$$

$$(-9.81 \text{ m/s}^2)(d)$$

$$(19.62 \text{ m/s}^2)(d) = (6.16 \text{ m/s})^2$$

$$d = \mathbf{1.93 \text{ m}}$$

No, she does not break the world record.



apex **1.93 m** at $t = \mathbf{0.628 \text{ s}}$ · record 2.09 m · **0.16 m short**

Two routes to the same answer: find the time to the apex and substitute it into the displacement equation, or go straight there with $v_f^2 = v_0^2 + 2ad$, which has no t in it at all.

Example Problem #3

A soccer player kicks a ball at an angle of 25° from horizontal, with an initial speed of 21 m/s. **How high and far does the ball go?**

First we will resolve the resultant velocity vector into its components. Then we will use the vertical velocity component to calculate the height that the ball reaches and the time that the ball spends in the air. Finally we will use this time value and the horizontal velocity component to calculate the horizontal distance.



Solution — resolve the vector

Vertical Component:

$$\sin\theta = \text{Opposite} / \text{Hypotenuse}$$

$$\text{Opposite} = (\text{Hypotenuse})(\sin\theta)$$

$$\text{Opposite} = (21 \text{ m/s})(\sin 25^\circ)$$

$$\text{Opposite} = \mathbf{8.9 \text{ m/s}}$$

Horizontal Component:

$$\cos\theta = \text{Adjacent} / \text{Hypotenuse}$$

$$\text{Adjacent} = (\text{Hypotenuse})(\cos\theta)$$

$$\text{Adjacent} = (21 \text{ m/s})(\cos 25^\circ)$$

$$\text{Adjacent} = \mathbf{19.0 \text{ m/s}}$$

21 m/s at 25° above horizontal

Solution — the path upward

Time: v_0 (vertical) = 8.9 m/s, v_f (vertical) = 0 m/s, $a = -9.81 \text{ m/s}^2$

$$v_f = at + v_0$$

$$(0 \text{ m/s}) = (8.9 \text{ m/s}) + (-9.81 \text{ m/s}^2)(t)$$

$$t = \mathbf{0.91 \text{ s}}$$

Remember that this is the time taken for the ball to travel from the ground to its highest point. For the way up and down, we must multiply this value by two. $\mathbf{0.91 \text{ s} \times 2 = 1.82 \text{ s}}$

Vertical displacement:

$$d(t)_y = \frac{1}{2}at^2 + v_{0y}t + d_{0y}$$

Or, with the equation from slide 23:

$$v_f^2 = v_0^2 + 2ad$$

$$(0 \text{ m/s})^2 = (8.9 \text{ m/s})^2 + (2)(-9.81 \text{ m/s}^2)(d)$$

$$d = \mathbf{4.04 \text{ m}}$$

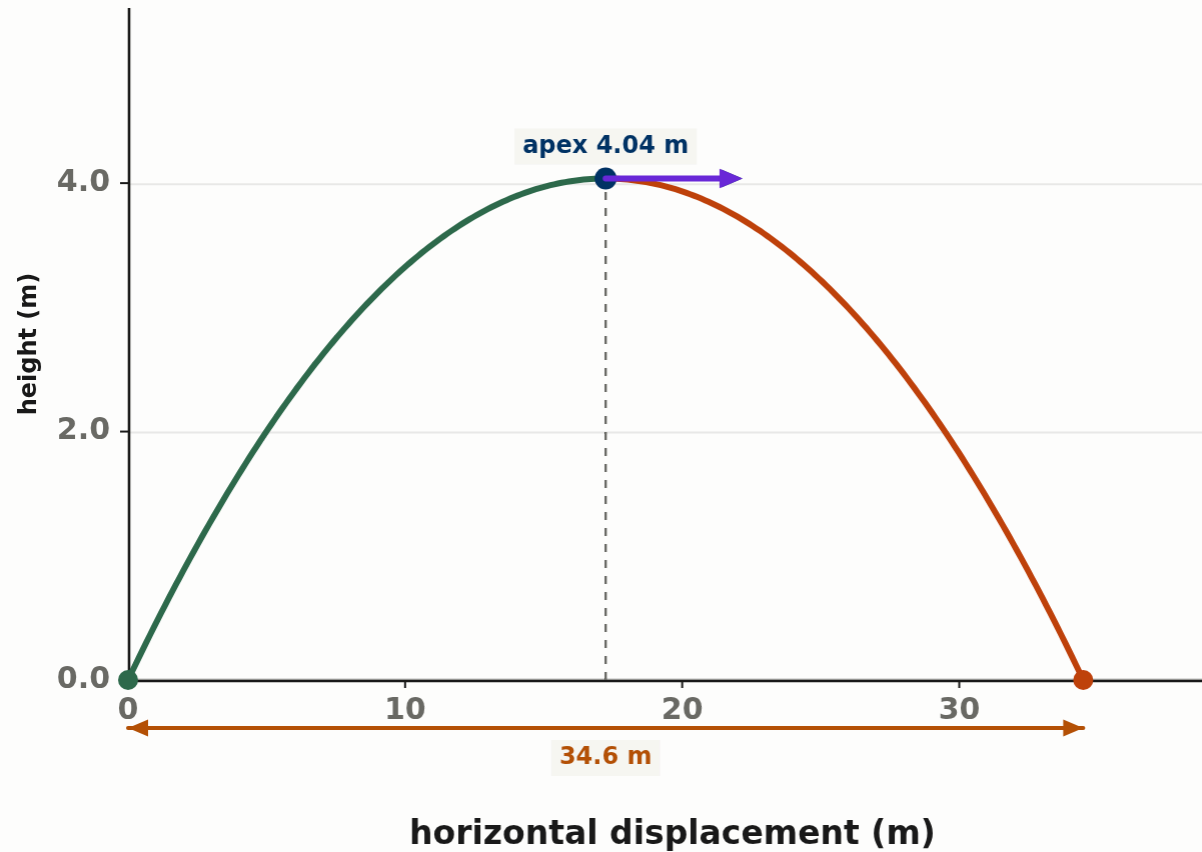
Horizontal distance travelled:

$$d = v_0t + d_0$$

$$d = (19.0 \text{ m/s})(1.82 \text{ s})$$

$$d = \mathbf{34.6 \text{ m}}$$

Example Problem #3 — the figure



Soccer ball, 21 m/s at 25°

- 1 · resolve the resultant velocity vector
- 2 · the path upward — time
- 3 · the path upward — vertical displacement

4 · the horizontal distance

$$d = v_0 t + d_0$$

$$d = (19.0 \text{ m/s})(1.82 \text{ s})$$

$d = \mathbf{34.6 \text{ m}}$ — the ball travels
34.6 m horizontally

v_x **19.0** · v_y **8.9 m/s** · time in the air **1.82 s** · apex **4.04 m** · distance **34.6 m**

Vertical and horizontal are solved separately and joined only by **time**. The vertical half gives how long the object is in the air; the horizontal half turns that time into a distance.

1 · resolve the resultant velocity vector

2 · the path upward — time

3 · the path upward — vertical displacement

Example Problem #4



Solution — convert and resolve

$$120 \text{ km/hr} \times 1000 \text{ m} \div 3600 \text{ s} = 33.3 \text{ m/s}$$

Vertical Component:

$$\text{Opposite} = (\text{Hypotenuse})(\sin\theta)$$

$$\text{Opposite} = (33.3 \text{ m/s})(\sin 37^\circ)$$

$$\text{Opposite} = \mathbf{20.0 \text{ m/s}}$$

Horizontal Component:

$$\cos\theta = \text{Adjacent} / \text{Hypotenuse}$$

$$\text{Adjacent} = (\text{Hypotenuse})(\cos\theta)$$

$$\text{Adjacent} = (33.3 \text{ m/s})(\cos 37^\circ)$$

$$\text{Adjacent} = \mathbf{26.6 \text{ m/s}}$$

33.3 m/s at 37° above horizontal, from 1.2 m

Solution — up, then down

The Path Upward:

$$v_f = at + v_0$$

$$(0 \text{ m/s}) = (-9.81 \text{ m/s}^2)(t) + (20.0 \text{ m/s})$$

$$t_{\text{up}} = \mathbf{2.04 \text{ s}}$$

Calculates the time for the ball to travel from the bat to the apex.

$$d = \left(\frac{1}{2}\right)at^2 + v_0t + d_0$$

$$d = \left(\frac{1}{2}\right)(-9.81 \text{ m/s}^2)(2.04)^2 + (20 \text{ m/s})(2.04) + (0 \text{ m or } 1.2 \text{ m})$$

$$d = \mathbf{20.4 \text{ m or } 21.6 \text{ m}}$$

The Path Downward: the vertical distance travelled is equal to that travelled during the path upward plus 1.2 m. The initial velocity is 0 m/s, but we do not know the final velocity.

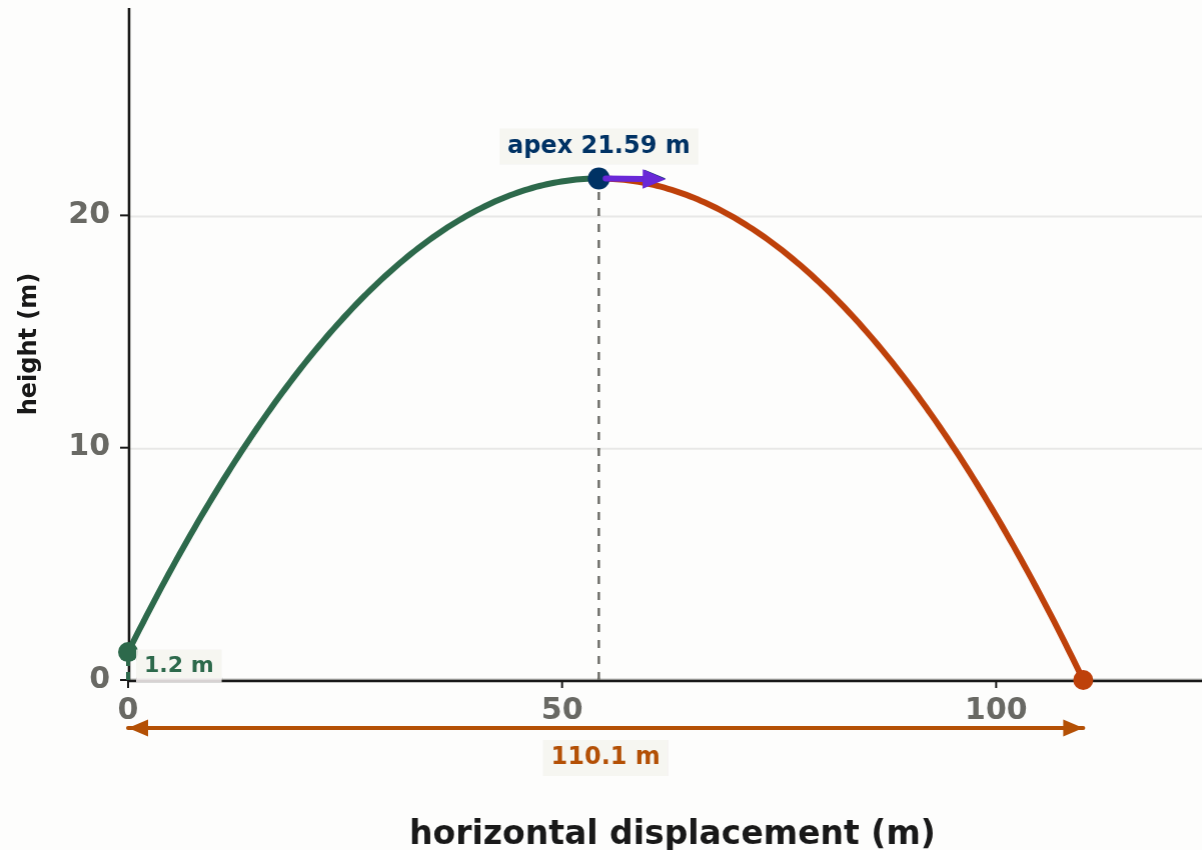
$$d = \left(\frac{1}{2}\right)at^2 + v_0t + d_0$$

$$0 \text{ m} = \left(\frac{1}{2}\right)(-9.81 \text{ m/s}^2)(t)^2 + (0 \text{ m/s})(t) + (20.4 \text{ m} + 1.2 \text{ m})$$

$$t_{\text{down}} = \mathbf{2.10 \text{ s}}$$

Solution — total time and distance

Example Problem #4 — the figure



Abreu's baseball, 120 km/hr at 37° from 1.2 m

1 · convert, then resolve

2 · the path upward

3 · the path downward

4 · total time, then the distance

$$t_{\text{total}} = 2.04 \text{ s} + 2.10 \text{ s} = \mathbf{4.14 \text{ s}}$$

$$d = \left(\frac{1}{2}\right)(0 \text{ m/s}^2)(t)^2 + (26.6 \text{ m/s})(4.14 \text{ s})$$

$$d = \mathbf{110.1 \text{ m}}$$

The ball reaches a height of 21.6 m and travels 110.1 m horizontally.

v_x **26.6** · v_y **20.0 m/s** · time in the air **4.14 s** · apex **21.6 m** · distance **110.1 m**

Vertical and horizontal are solved separately and joined only by **time**. The vertical half gives how long the object is in the air; the horizontal half turns that time into a distance.

1 · convert, then resolve

2 · the path upward

3 · the path downward

4 · total time, then the distance

Sample Problem

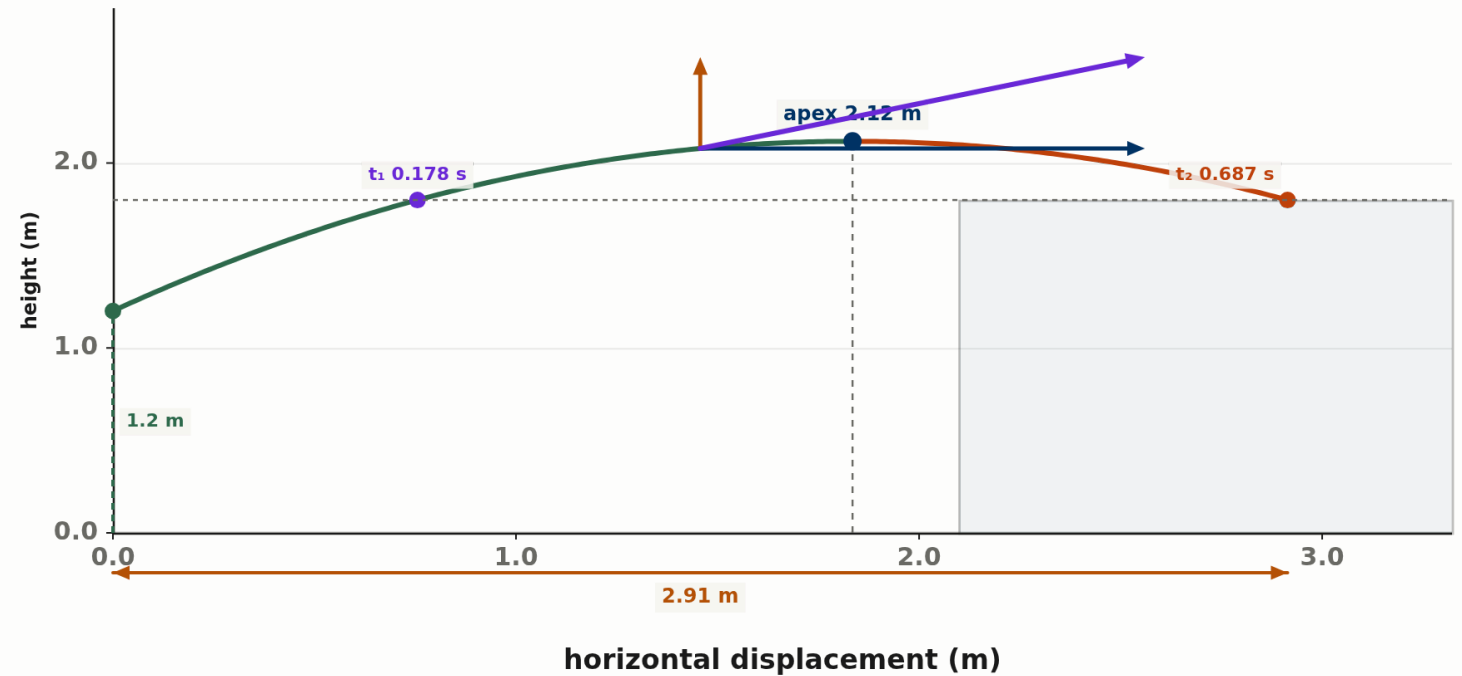
Find the time in the air for a man jumping up on the platform and how far he will travel horizontally

Given:

initial velocity = 6 m/s at 45°

height of landing = 1.8 m

The jumper leaves the ground at a hip height of 1.2 m and lands on a platform at 1.8 m.



v_x 4.2 · v_y 4.2 m/s · time in the air 0.687 s · apex 2.12 m · distance 2.91 m

The landing height is reached **twice**, so the quadratic has two roots and only the later one is the landing. Substituting the earlier one into the velocity equation gives a positive value — the jumper is still on the way up.

Step 1: Derive the equations of motion

Since gravity is the only force acting on the subject, the acceleration in the y direction is -9.81 m/s^2 and the acceleration in the x direction is 0 m/s^2 . We use integral calculus to yield the velocity and displacement equations.

$$1) a_y(t) = -9.81$$

$$2) v_y(t) = -9.81t + v_{oy}$$

$$3) y(t) = (-9.81/2)t^2 + v_{oy}t + y_o$$

$$4) a_x(t) = 0$$

$$5) v_x(t) = v_{ox}$$

$$6) x(t) = v_{ox}t + x_o$$

Step 2: Substitute known values for variables

Now we substitute the given information of this particular problem.

$$1) a_y(t) = -9.81$$

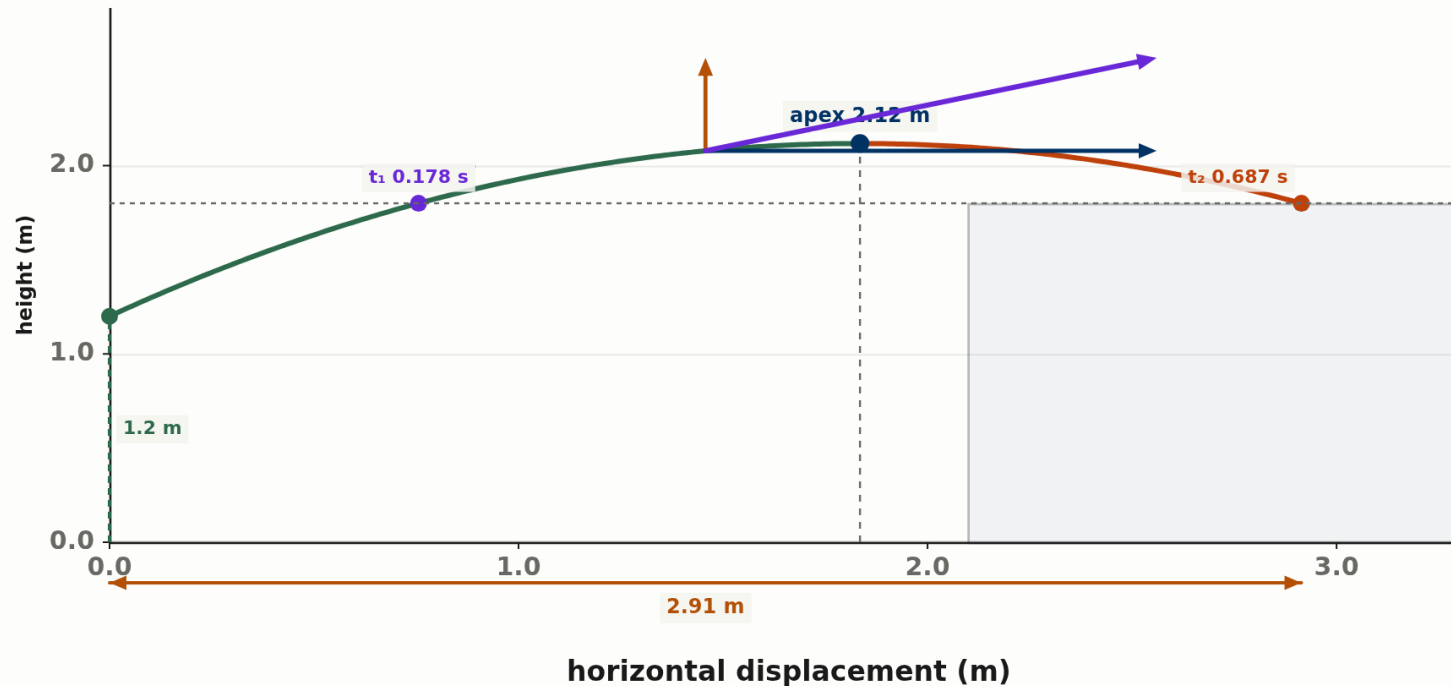
$$2) v_y(t) = -9.81t + 6\sin 45^\circ$$

$$3) y(t) = (-9.81/2)t^2 + 6\sin 45^\circ t + 1.2$$

$$4) a_x(t) = 0$$

$$5) v_x(t) = 6\cos 45^\circ$$

$$6) x(t) = 6\cos 45^\circ t + 0$$



Step 3: Use the appropriate equations

This is the most difficult step because we must look at our six equations of motion and decide which one or ones will help us find the answer to the question that we have been asked. In this case we have been asked to find the time in the air. Although time is part of all of the equations, we know that at the instant of landing, the vertical displacement is 1.8 meters. Equation [3] is the vertical displacement equation and we must use this equation to solve for t when $y(t) = 1.8$.

$$y(t) = (-9.81/2)t^2 + 6\sin 45^\circ t + 1.2$$

$$1.8 = (-9.81/2)t^2 + 6\sin 45^\circ t + 1.2$$

$$0 = (-9.81/2)t^2 + 6\sin 45^\circ t - 0.6 \quad \textbf{This is a quadratic equation}$$

Step 4: Solve for the unknown variables

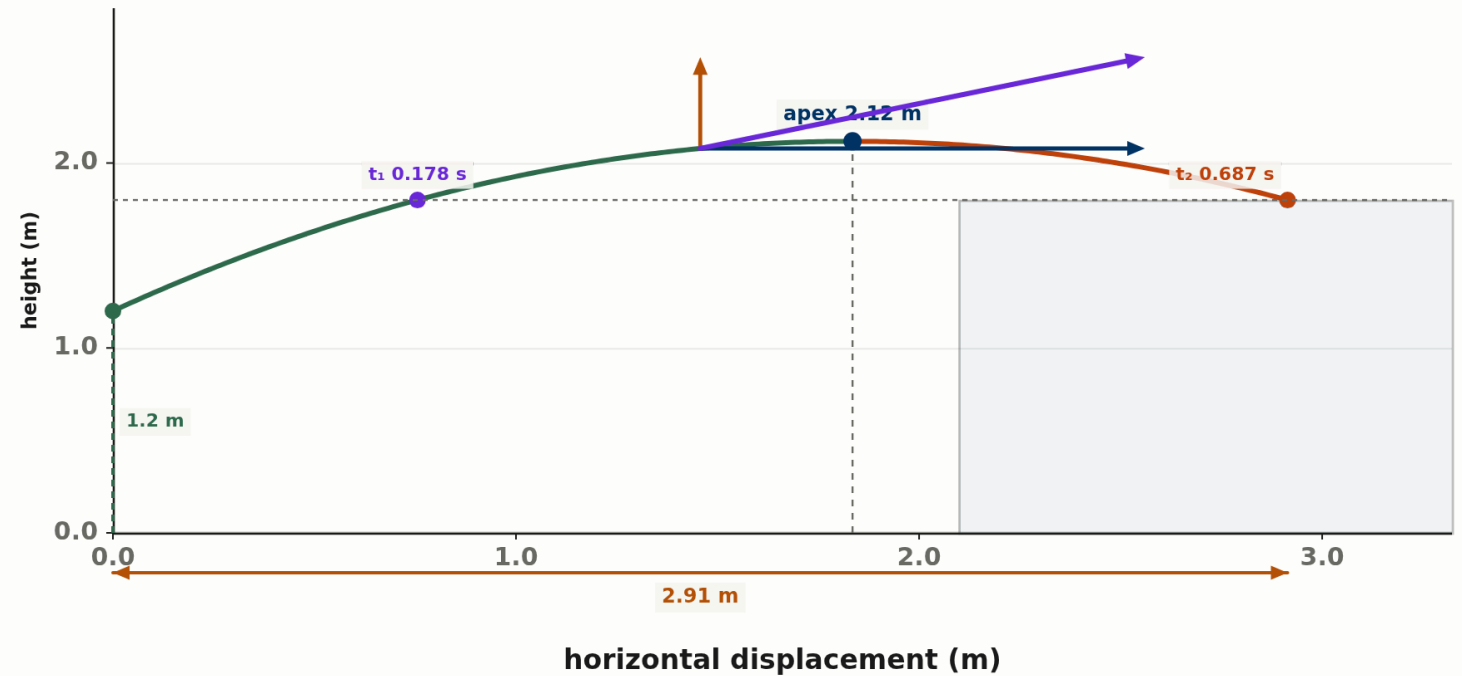
We must use the quadratic formula to solve for the two values of t which satisfy this equation. Root of a Quadratic Equation for an equation in the form $\mathbf{a \cdot x^2 + b \cdot x + c = 0}$, solve for x :

$$x = \frac{-b \pm \sqrt{b^2 - 4 \cdot a \cdot c}}{2 \cdot a}$$

In this case $a = -4.905$, $b = 6\sin 45^\circ$, $c = -0.6$

Solving the equation above leaves two possible answers for t (using the positive and negative roots). Therefore we are left with two values of t :

$t = 0.178$ and $t = 0.687$



v_x **4.2** · v_y **4.2 m/s** · time in the air **0.687 s** · apex **2.12 m** · distance **2.91 m**

The landing height is reached **twice**, so the quadratic has two roots and only the later one is the landing. Substituting the earlier one into the velocity equation gives a positive value — the jumper is still on the way up.

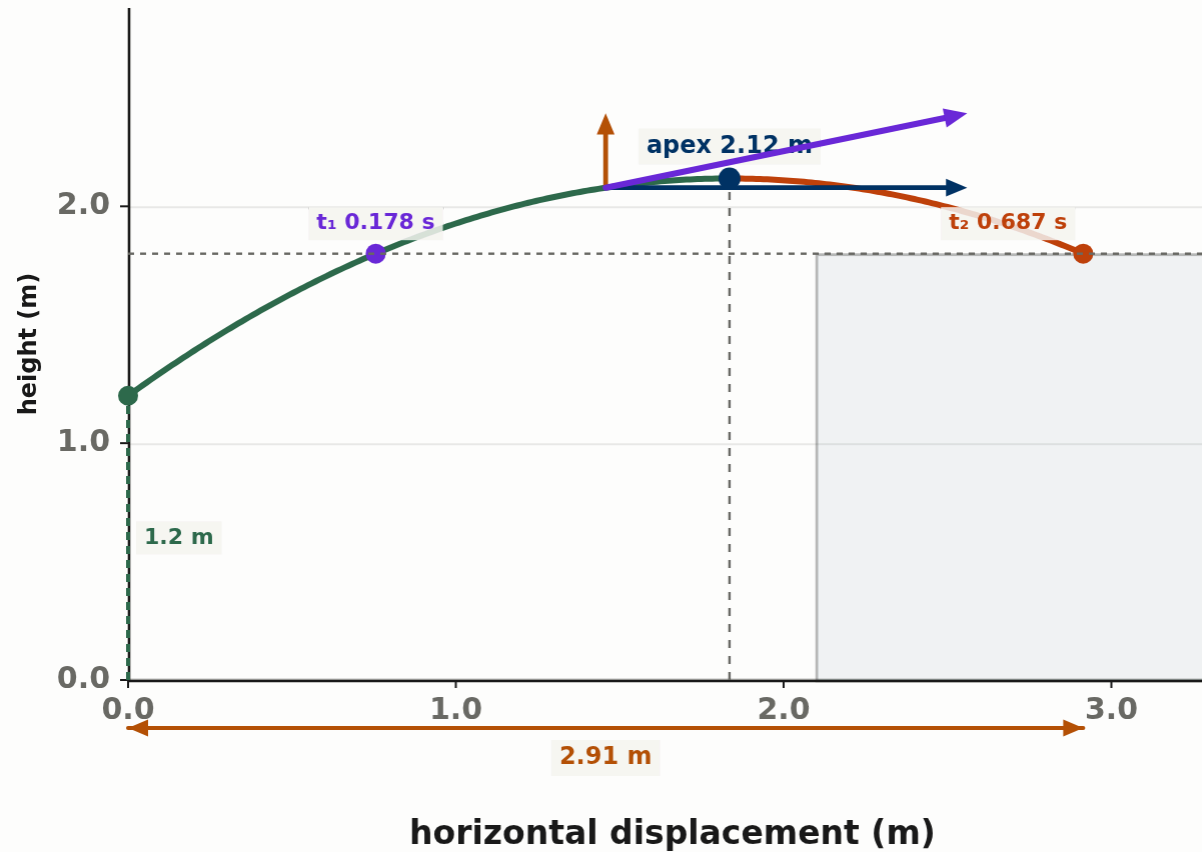
Which root is the landing?

Due to the parabolic path of the jumper, the jumper's height was 1.8 meters high at two instants in time (t_1 and t_2). We want the later time (t_2) because that is the time of landing. At the earlier time (t_1) the jumper was travelling upward. If we are ever uncertain about the direction, we can substitute the time into equation [2] and if velocity is positive, the motion is upward and if the velocity is negative, the direction is downward. **Therefore, the time in the air is 0.687 seconds.**

We were also asked to calculate the horizontal displacement on landing. We can use equation [6] which is the horizontal displacement function knowing the time of landing is 0.687 seconds.

$$\begin{aligned}x(t) &= 6\cos 45^\circ t + 0 \\&= 6\cos 45^\circ (0.687) + 0 \\&= \mathbf{2.91 \text{ m}}\end{aligned}$$

Sample Problem — the whole thing



Jumping onto a platform, 6 m/s at 45°

- 1 · derive the equations of motion
- 2 · substitute the known values
- 3 · use the appropriate equation
- 4 · solve for the two roots

5 · which root, and how far
the jumper passes 1.8 m twice —
on the way up at 0.178 s and on
the way down at 0.687 s
landing is the later one, so the
time in the air is **0.687 s**
 $x(t) = 6\cos 45^\circ(0.687) + 0 = \mathbf{2.91}$
m

v_x **4.2** · v_y **4.2 m/s** · time in the air **0.687 s** · apex **2.12 m** · distance **2.91 m**

The landing height is reached **twice**, so the quadratic has two roots and only the later one is the landing. Substituting the earlier one into the velocity equation gives a positive value — the jumper is still on the way up.

1 · derive the equations of motion

2 · substitute the known values

3 · use the appropriate equation

4 · solve for the two roots

Summary

- A projectile moves under gravity alone, so $a_y = -9.81 \text{ m/s}^2$ and $a_x = 0$
- Integrating those twice gives the six equations of motion, and the constants are the initial velocity and the initial position
- Vertical and horizontal are solved **separately** and share only **time**
- At the apex $v_y = 0$; with a level take-off and landing the apex is at half the flight time
- 45° is best for distance from ground level, 90° for height — raise the release point and the best angle for distance drops below 45°
- When take-off and landing heights differ, the flight time comes from a **quadratic**.

